

Environmental Monitoring Report

Report RPT-261005-4F051D · Monitoring location: Kondapalli, Vijayawada · Period 2026-09-28 05:02 to 2026-10-05 05:02

Human-review status

APPROVED by Environmental Officer (demo) at 2026-10-05 05:02

Monitoring stations

Station	Name	Type	Location
ENV-ST-001	Benz Circle AQ Station	air quality	Benz Circle, Vijayawada
ENV-ST-002	Auto Nagar AQ Station	air quality	Auto Nagar, Vijayawada
ENV-ST-003	Gunadala AQ Station	air quality	Gunadala, Vijayawada
ENV-ST-004	Industrial Zone A AQ Station	air quality	Industrial Zone A, Kondapalli
ENV-WQ-001	Krishna River - Prakasam Barrage	water quality	Prakasam Barrage, Vijayawada
ENV-WQ-002	Budameru Canal Outfall	water quality	Budameru Canal, Vijayawada
ENV-NS-001	Governorpet Noise Station	noise	Governorpet, Vijayawada
ENV-NS-002	Patamata Noise Station	noise	Patamata, Vijayawada

Air-quality summary

ENV-ST-001 — Benz Circle AQ Station: Investigation Required (risk index 60.0)

Benz Circle AQ Station: Investigation Required. PM2.5 24-h mean above configured 24h reference. Monitoring risk index 60.0 (HIGH). Suggested next step for officer review: Dispatch field inspection to the monitoring area. Findings support investigation and do not constitute a legal determination.

ENV-ST-002 — Auto Nagar AQ Station: Critical Review Required (risk index 65.3)

Auto Nagar AQ Station: Critical Review Required. PM2.5 24-h mean above configured 24h reference. Monitoring risk index 65.3 (HIGH). Suggested next step for officer review: Increase monitoring frequency for 48 h. Findings support investigation and do not constitute a legal determination.

ENV-ST-003 — Gunadala AQ Station: Sensor Verification Required (risk index 29.5)

Gunadala AQ Station: Sensor Verification Required. Unrealistic co reading(s) flagged — sensor verification requested. Monitoring risk index 29.5 (MODERATE). Suggested next step for officer review: Dispatch technician to inspect sensor. Findings support investigation and do not constitute a legal determination.

ENV-ST-004 — Industrial Zone A AQ Station: Critical Review Required (risk index 66.4)

Industrial Zone A AQ Station: Critical Review Required. PM2.5 24-h mean above configured 24h reference. Monitoring risk index 66.4 (HIGH). Suggested next step for officer review: Prepare public health advisory draft for approval. Findings support investigation and do not constitute a legal determination.

Water-quality summary

ENV-WQ-001 — Krishna River - Prakasam Barrage: Investigation Required (risk index 61.6)

Krishna River - Prakasam Barrage: Investigation Required. Turbidity latest instantaneous reading above configured instantaneous reference. Monitoring risk index 61.6 (HIGH). Suggested next step for officer review: Inspect upstream outfalls (sources requiring investigation). Findings support investigation and do not constitute a legal determination.

ENV-WQ-002 — Budameru Canal Outfall: Sensor Verification Required (risk index 34.8)

Budameru Canal Outfall: Sensor Verification Required. ENV-WQ-002 not reporting — stale sensor. Monitoring risk index 34.8 (MODERATE). Suggested next step for officer review: Mark readings invalid pending verification. Findings support investigation and do not constitute a legal determination.

Noise summary

ENV-NS-001 — Governorpet Noise Station: Investigation Required (risk index 53.7)

Governorpet Noise Station: Investigation Required. Repeated night-time noise above ≤ 55 dB(A) (7 of 7 nights). Monitoring risk index 53.7 (HIGH). Suggested next step for officer review: Check complaints register for the area. Findings support investigation and do not constitute a legal determination.

ENV-NS-002 — Patamata Noise Station: Investigation Required (risk index 79.8)

Patamata Noise Station: Investigation Required. Repeated night-time noise above ≤ 45 dB(A) (7 of 7 nights). Monitoring risk index 79.8 (CRITICAL). Suggested next step for officer review: Review nearby venues (sources requiring investigation). Findings support investigation and do not constitute a legal determination.

Weather conditions

Source: dataset-fallback (Open-Meteo unreachable) · observed 2026-10-05T05:00:00 · 31.0 °C, 68.0 % RH, wind 4.5 m/s from 226.0°, rain 0.0 mm

- Elevated pm2_5 values in the last 24 h occurred during low-wind conditions (mean 1.31 m/s vs. 30-day mean 3.33 m/s).
- Over 30 days, pm2_5 and wind speed show a negative association (Spearman $\rho = -0.81$). Association does not establish causation.
- Weather is changing: wind speed moved from 1.31 m/s during the elevated period to 4.07 m/s in the last 3 h.

Environmental-standard comparisons

Measured value, applicable reference and calculated difference are deterministic; averaging periods match the standard.

Station	Parameter	Measured	Reference	Difference	%	Status
ENV-ST-001	PM2.5	83.456 (24-h mean)	$\leq 60 \mu\text{g}/\text{m}^3$ [24h]	23.456	39.1	exceedance
ENV-ST-001	PM10	134.418 (24-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [24h]	34.418	34.4	exceedance
ENV-ST-001	NO ₂	66.173 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-13.827	-17.3	compliant
ENV-ST-001	SO ₂	20.608 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-59.392	-74.2	compliant
ENV-ST-001	CO	1.051 (8-h mean)	$\leq 2 \text{ mg}/\text{m}^3$ [8h]	-0.949	-47.5	compliant
ENV-ST-001	O ₃	31.693 (8-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [8h]	-68.307	-68.3	compliant
ENV-ST-002	PM2.5	97.831 (24-h mean)	$\leq 60 \mu\text{g}/\text{m}^3$ [24h]	37.831	63.1	exceedance
ENV-ST-002	PM10	159.734 (24-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [24h]	59.734	59.7	exceedance
ENV-ST-002	NO ₂	78.391 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-1.609	-2.0	compliant
ENV-ST-002	SO ₂	28.69 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-51.31	-64.1	compliant
ENV-ST-002	CO	1.199 (8-h mean)	$\leq 2 \text{ mg}/\text{m}^3$ [8h]	-0.801	-40.1	compliant
ENV-ST-002	O ₃	29.119 (8-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [8h]	-70.881	-70.9	compliant
ENV-ST-003	PM2.5	34.442 (24-h mean)	$\leq 60 \mu\text{g}/\text{m}^3$ [24h]	-25.558	-42.6	compliant
ENV-ST-003	PM10	55.937 (24-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [24h]	-44.063	-44.1	compliant
ENV-ST-003	NO ₂	26.566 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-53.434	-66.8	compliant
ENV-ST-003	SO ₂	8.906 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-71.094	-88.9	compliant
ENV-ST-003	CO	0.635 (8-h mean)	$\leq 2 \text{ mg}/\text{m}^3$ [8h]	-1.365	-68.3	compliant
ENV-ST-003	O ₃	28.742 (8-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [8h]	-71.257	-71.3	compliant
ENV-ST-004	PM2.5	99.1 (24-h mean)	$\leq 60 \mu\text{g}/\text{m}^3$ [24h]	39.1	65.2	exceedance
ENV-ST-004	PM10	160.383 (24-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [24h]	60.383	60.4	exceedance
ENV-ST-004	NO ₂	96.292 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	16.292	20.4	exceedance
ENV-ST-004	SO ₂	39.883 (24-h mean)	$\leq 80 \mu\text{g}/\text{m}^3$ [24h]	-40.117	-50.1	compliant
ENV-ST-004	CO	1.17 (8-h mean)	$\leq 2 \text{ mg}/\text{m}^3$ [8h]	-0.83	-41.5	compliant
ENV-ST-004	O ₃	29.865 (8-h mean)	$\leq 100 \mu\text{g}/\text{m}^3$ [8h]	-70.135	-70.1	compliant
ENV-WQ-001	pH	7.474 (latest instantaneous reading)	6.0–9.0 pH [instantaneous]	-1.474	-24.6	compliant
ENV-WQ-001	Dissolved oxygen	5.2 (latest instantaneous reading)	$\geq 4.0 \text{ mg}/\text{L}$ [instantaneous]	-1.2	-30.0	compliant
ENV-WQ-001	Turbidity	12.8 (latest instantaneous reading)	$\leq 5 \text{ NTU}$ [instantaneous]	7.8	156.0	exceedance

Station	Parameter	Measured	Reference	Difference	%	Status
ENV-WQ-001	Conductivity	537.974 (latest instantaneous reading)	150-900 µS/cm [instantaneous]	-362.026	-40.2	compliant
ENV-WQ-001	TDS	345.162 (latest instantaneous reading)	≤ 2000 mg/L [instantaneous]	-1654.838	-82.7	compliant
ENV-WQ-002	pH	7.346 (latest instantaneous reading)	6.0-9.0 pH [instantaneous]	-1.346	-22.4	compliant
ENV-WQ-002	Dissolved oxygen	5.816 (latest instantaneous reading)	≥ 4.0 mg/L [instantaneous]	-1.816	-45.4	compliant
ENV-WQ-002	Turbidity	4.565 (latest instantaneous reading)	≤ 5 NTU [instantaneous]	-0.435	-8.7	compliant
ENV-WQ-002	Conductivity	605.156 (latest instantaneous reading)	150-900 µS/cm [instantaneous]	-294.844	-32.8	compliant
ENV-WQ-002	TDS	387.975 (latest instantaneous reading)	≤ 2000 mg/L [instantaneous]	-1612.025	-80.6	compliant
ENV-NS-001	Noise LAeq	56.5 (night-time Leq (2026-10-04))	≤ 55 dB(A) [night]	1.5	2.7	exceedance
ENV-NS-002	Noise LAeq	54.9 (night-time Leq (2026-10-04))	≤ 45 dB(A) [night]	9.9	22.0	exceedance

Detected anomalies

ENV-ST-001: 22 anomalous hour(s) in the last 24 h (max score 1.0).

ENV-ST-002: 22 anomalous hour(s) in the last 24 h (max score 1.0).

ENV-ST-003: 1 reading(s) flagged for sensor verification.

ENV-ST-004: 23 anomalous hour(s) in the last 24 h (max score 1.0).

ENV-WQ-001: 6 anomalous hour(s) in the last 24 h (max score 1.0). Unusual multi-parameter pattern: turbidity, conductivity, tds. turbidity change classified as valid environmental change. conductivity change classified as valid environmental change. tds change classified as valid environmental change.

ENV-NS-002: 4 anomalous hour(s) in the last 24 h (max score 0.993).

Domain	Isolation Forest P / R / F1	z-score baseline P / R / F1
air	0.796 / 0.932 / 0.859	0.764 / 0.92 / 0.835
water	0.545 / 1.0 / 0.706	1.0 / 1.0 / 1.0
noise	0.424 / 0.583 / 0.491	0.0 / 0.0 / 0.0

Pollution trends & forecast

ENV-ST-001: pm2_5 increasing, pm10 increasing, no2 increasing, so2 increasing, co increasing, o3 stable

ENV-ST-002: pm2_5 increasing, pm10 increasing, no2 increasing, so2 increasing, co increasing, o3 stable

ENV-ST-003: pm2_5 stable, pm10 stable, no2 stable, so2 stable, co stable, o3 stable

ENV-ST-004: pm2_5 increasing, pm10 increasing, no2 increasing, so2 increasing, co increasing, o3 stable

PM2.5 24-h forecast for ENV-ST-001 (HistGradientBoostingRegressor (recursive multi-step)): mean 83.4, range 49.92-101.05 µg/m³. One-step MAE 9.509, RMSE 16.522, MAPE 13.84 % (persistence MAE 13.89). 24-h backtest MAE 10.179 vs persistence 14.568. Future weather: diurnal persistence (same hour previous day) - live forecast unavailable.

Environmental incidents & investigation findings

INC-261005-2DA324 · ENV-NS-002 · noise_laeq · Investigation Required · status open (awaiting_review)

Repeated night-time noise above ≤ 45 dB(A) (7 of 7 nights)

2026-10-05 05:02 — created by Alert Agent: Repeated night-time noise above ≤ 45 dB(A) (7 of 7 nights)

INC-261005-5FD8D8 · ENV-NS-001 · noise_laeq · Investigation Required · status open (awaiting_review)

Repeated night-time noise above ≤ 55 dB(A) (7 of 7 nights)

2026-10-05 05:02 — created by Alert Agent: Repeated night-time noise above ≤ 55 dB(A) (7 of 7 nights)

INC-261005-87762C · ENV-WQ-002 · * · Sensor Verification Required · status open (awaiting_review)

ENV-WQ-002 not reporting — stale sensor

2026-10-05 05:02 — created by Alert Agent: ENV-WQ-002 not reporting — stale sensor

INC-261005-D68EDD · ENV-WQ-001 · turbidity · Investigation Required · status open (awaiting_review)

Turbidity latest instantaneous reading above configured instantaneous reference

2026-10-05 05:02 — created by Alert Agent: Turbidity latest instantaneous reading above configured instantaneous reference

INC-261005-50EC2A · ENV-ST-003 · co · Sensor Verification Required · status open (awaiting_review)

Unrealistic co reading(s) flagged — sensor verification requested

2026-10-05 05:02 — created by Alert Agent: Unrealistic co reading(s) flagged — sensor verification requested

INC-261005-B7DDA9 · ENV-ST-001 · pm10,pm2_5 · Investigation Required · status open (awaiting_review)

PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — created by Alert Agent: PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-F88AED: PM10 24-h mean above configured 24h reference

INC-261005-426FF9 · ENV-ST-002 · pm10,pm2_5 · Critical Review Required · status open (awaiting_review)

PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — created by Alert Agent: PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-0B5F12: PM10 24-h mean above configured 24h reference

INC-261005-8FF52E · ENV-ST-004 · no2,pm10,pm2_5 · Critical Review Required · status open (awaiting_review)

PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — created by Alert Agent: PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-051660: PM10 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-051660: NO₂ 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-68C872: PM2.5 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-68C872: PM10 24-h mean above configured 24h reference

2026-10-05 05:02 — evidence_updated by Alert Agent: Run RUN-261005-68C872: NO₂ 24-h mean above configured 24h reference

Recommended actions (for officer decision)

- Check complaints register for the area
- Dispatch field inspection to the monitoring area
- Dispatch technician to inspect sensor
- Increase monitoring frequency for 48 h
- Inspect upstream outfalls (sources requiring investigation)
- Mark readings invalid pending verification

- Request PM analyser flow/tape check
- Review nearby venues (sources requiring investigation)

Source references

Standard ID	Standard	Section	Document	Version	Basis
IN-NAAQS-PM25-24H	National Ambient Air Quality Standards (NAAQS) 2009	Schedule, item 4 - PM2.5	naaqs_2009.md	Notification G.S.R. 826(E), November 2009	regulatory
IN-NAAQS-PM10-24H	National Ambient Air Quality Standards (NAAQS) 2009	Schedule, item 3 - PM10	naaqs_2009.md	Notification G.S.R. 826(E), November 2009	regulatory
IN-NAAQS-NO2-24H	National Ambient Air Quality Standards (NAAQS) 2009	Schedule, item 2 - NO2	naaqs_2009.md	Notification G.S.R. 826(E), November 2009	regulatory
IN-NAAQS-SO2-24H	National Ambient Air Quality Standards (NAAQS) 2009	Schedule, item 1 - SO2	naaqs_2009.md	Notification G.S.R. 826(E), November 2009	regulatory
IN-NAAQS-CO-8H	National Ambient Air Quality Standards (NAAQS) 2009	Schedule, item 7 - CO	naaqs_2009.md	Notification G.S.R. 826(E), November 2009	regulatory
IN-NAAQS-O3-8H	National Ambient Air Quality Standards (NAAQS) 2009	Schedule, item 5 - Ozone	naaqs_2009.md	Notification G.S.R. 826(E), November 2009	regulatory
IN-CPCB-DBU-C-PH	CPCB Primary Water Quality Criteria (Designated Best Use)	Class C - Drinking water source after conventional treatment	cpcb_designate_d_best_use.md	CPCB criteria (current listing)	regulatory
IN-CPCB-DBU-C-D O	CPCB Primary Water Quality Criteria (Designated Best Use)	Class C - Drinking water source after conventional treatment	cpcb_designate_d_best_use.md	CPCB criteria (current listing)	regulatory
IN-IS10500-TURB-SCREEN	IS 10500:2012 Drinking Water Specification (screening benchmark)	Table 1 - Turbidity, permissible limit	is_10500_2012.md	IS 10500:2012 (2nd revision)	screening_benchmark
SITE-WQ-COND-BASELINE	Site operating baseline - conductivity	Conductivity operating band	site_baseline_config.md	v1.2, 2026	site_baseline
IN-IS10500-TDS-SCREEN	IS 10500:2012 Drinking Water Specification (screening benchmark)	Table 1 - Total dissolved solids, permissible limit	is_10500_2012.md	IS 10500:2012 (2nd revision)	screening_benchmark
IN-NOISE-COM-NIGHT	Noise Pollution (Regulation and Control) Rules, 2000	Schedule - Ambient air quality standards in respect of noise (Category B - Commercial area)	noise_rules_2000.md	S.O. 123(E), 2000 (as amended)	regulatory
IN-NOISE-RES-NIGHT	Noise Pollution (Regulation and Control) Rules, 2000	Schedule - Ambient air quality standards in respect of noise (Category C - Residential area)	noise_rules_2000.md	S.O. 123(E), 2000 (as amended)	regulatory

Decision-support output. Comparisons use configured standards; findings do not constitute legal determinations, and no organisation is identified as the cause of any event without independent confirmation.